Do LLMs Know When They're Making Things Up?

Hallucination detection on HaluEval QA

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Abstract

This report evaluates three approaches to detecting hallucinated answers on 200 questions from the QA subset of HaluEval, using the open model Qwen2.5-3B-Instruct. Each question is paired with a correct and a hallucinated answer. Token log-probability and entropy (Task 1) perform below chance. SelfCheck-NLI (Task 2) attains an AUC-PR of 0.65 and SelfCheck-BERTScore 0.32. A logistic regression over all four scores (Task 3) attains 0.95 on held-out questions, but answer length alone attains 0.97, since hallucinated answers are about four times longer than correct ones. After controlling for length, the combined scores retain an AUC-PR of 0.64.

Code. The code, results and instructions are available at

`https://github.com/susanta75/NLP-Project-2026-Autumn`

To run all tasks with Docker:

`bash run_docker.sh`	Tasks 1–3 from the saved model outputs
`bash run_docker.sh --recompute`	additionally reruns the GPU steps of Tasks 1 and 2

Without Docker: `pip install -r requirements.txt`, then `bash code/run_all.sh`. Each task can also be run independently (`python code/taskN/taskN.py`). The README lists the data and model sources and the location of every output.

1 Introduction

A hallucination is a fluent and confident response that is factually incorrect. Because such a response is not distinguishable from a correct one on the surface, detection must rely on other signals. Three signals are examined: the model's own token probabilities (Task 1), consistency with stochastic samples from the same model, following SelfCheckGPT (Task 2), and a logistic regression combining both (Task 3). The evaluation uses the QA subset of HaluEval, in which each HotpotQA question is paired with a correct answer and a hallucinated answer generated by ChatGPT. The two answer types differ systematically in form: correct answers are short (e.g. *"no"*, *"Lanoxin"*), whereas hallucinated answers are typically complete sentences. This property affects all subsequent results.

2 Experimental setup

- **Data.** 200 of the 10,000 HaluEval QA pairs, sampled with seed 42. Each pair yields two items: the correct answer (label 0) and the hallucinated answer (label 1).
- **Model.** `Qwen/Qwen2.5-3B-Instruct` in float16 on a single RTX 5060 Ti (16 GB). The prompt is a system instruction (*"Answer the question concisely and factually."*) followed by the question. HaluEval's `knowledge` paragraph is not provided (closed-book setting).

- **Metric.** Each detector assigns every answer a score, where a higher score indicates a more likely hallucination. Ranking answers by score gives precision P_k and recall R_k at each cut-off k, and

$$\text{AUC-PR} = \sum_k (R_k - R_{k-1})\, P_k. \tag{1}$$

  Since half of the answers are hallucinated, a random detector attains approximately 0.5. AUROC and the fraction of pairs in which the hallucinated answer receives the higher score ("pairs ranked correctly") are also reported.

- **Length baseline.** Correct answers have a median length of 3 tokens (mean 3.7, range 1–13); hallucinated answers have a median of 14.5 (mean 16.1, range 2–46). The hallucinated answer is the longer one in 194 of 200 pairs, and ranking by length alone gives an AUC-PR of 0.972. Length is therefore reported as a reference baseline next to every detector.

- **Random seeds.** Question sample: 42. Task 2 sampling: 42 + question id. Task 3 split and logistic regression: 42. Bootstrap intervals: 42. Repeated splits: 0–99.

3 Method summary

3.1 Task 1: Token probabilities

- The prompt x and an answer $a = (a_1, \ldots, a_T)$ are passed through the model in a single forward pass (forced decoding). No text is generated.

- Two quantities are averaged over the answer positions:

$$\bar{\ell}(a) = \frac{1}{T}\sum_{t=1}^{T} \log p(a_t \mid x, a_{<t}), \qquad \bar{H}(a) = \frac{1}{T}\sum_{t=1}^{T} \Big(-\sum_v p(v \mid x, a_{<t}) \log p(v \mid x, a_{<t}) \Big), \tag{2}$$

  where the inner sum runs over the vocabulary.

- The hallucination scores are $-\bar{\ell}$ and $\bar{H}$. The minimum token log-probability and the maximum token entropy are reported in addition, to test whether averaging hides isolated uncertain tokens.

3.2 Task 2: SelfCheckGPT

- HaluEval provides no samples, so $N = 20$ samples were generated per question with the same model and prompt (temperature 1.0, top-$p = 1$, no top-k, at most 48 new tokens, seed 42 + question id). Both answers of a pair are compared with the same samples.

- Scoring uses the official `selfcheckgpt` package (version 0.1.7). For an answer r and samples $S^1, \ldots, S^N$, with s_k^n the k-th sentence of S^n,

$$S_{\text{BERT}}(r) = 1 - \frac{1}{N}\sum_{n=1}^{N} \max_k \text{ BERTScore-F1}\big(r,\, s_k^n\big), \tag{3}$$

$$S_{\text{NLI}}(r) = \frac{1}{N}\sum_{n=1}^{N} P\big(\text{contradiction} \mid r,\, S^n\big). \tag{4}$$

- BERTScore uses `roberta-large` (layer 17) with baseline rescaling. The NLI model is DeBERTa-v3-large fine-tuned on MNLI. No model is trained in this task.

3.3 Task 3: Stacked logistic regression

- The 200 questions were split 80/20 with seed 42 (320 training and 80 test answers). The split is made by question, so both answers of a question fall in the same partition.

- Each answer is represented by the vector z of its four scores ($\bar{\ell}$, $\bar{H}$, S_{BERT}, S_{NLI}), standardised on the training set. A logistic regression (L2, $C = 1$) is fitted:

$$P(\text{hallucinated} \mid z) = \sigma(w^\top z + b) = \frac{1}{1 + e^{-(w^\top z + b)}}. \tag{5}$$

- Tasks 1, 2 and 3 are evaluated on the same 80 test answers; Tasks 1 and 2 also as a logistic regression on their own two scores. 95% confidence intervals use 1000 bootstrap resamples of the 40 test questions.

3.4 Deviations from the specification

- **Model size.** A 3B model was used so that forced decoding, sampling and scoring all fit on one 16 GB GPU.
- **Sample size.** 200 of the 10,000 questions were used, because Task 2 requires 20 generated samples per question (4,000 generations in total).
- **NLI model.** `microsoft/deberta-v3-large-mnli` does not exist on Hugging Face. The model loaded by the official code, `potsawee/deberta-v3-large-mnli`, is used instead.
- **Library compatibility.** `selfcheckgpt` 0.1.7 calls a function removed in transformers 5. A patch in `task2.py` restores it; the package then reproduces the example scores in its README to six decimal places.
- **Missing BERTScore values.** BERTScore ignores sample sentences of three tokens or fewer. For two questions all samples were single words, so four answers have no BERTScore ($n = 396$ in Task 2). In Task 3 these values are imputed with the training median.
- **Split by question.** The specification asks for an 80/20 split of items. Splitting by question prevents the classifier from being trained on one answer and tested on the other answer to the same question.

4 Quantitative analysis

4.1 Task 1

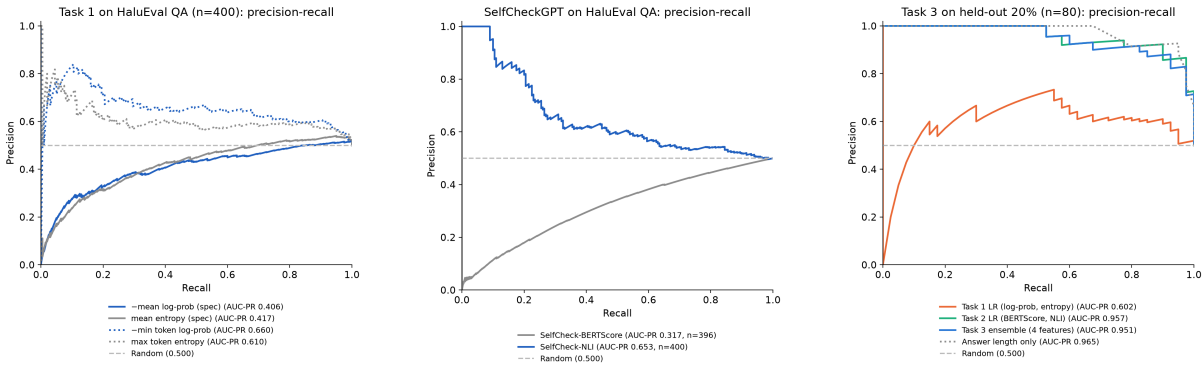
- Both specified scores perform below chance (Table 1, Figure 1a).
- The hallucinated answer receives the higher mean log-probability in 63% of pairs (37% ranked correctly); the median $\bar{\ell}$ is -3.62 for correct answers and -2.49 for hallucinated ones.
- The minimum log-probability (0.66) and maximum entropy (0.61) perform above chance, but a longer answer has more opportunities to contain a rare token, so both depend on length.

Task 1 score	AUC-PR	AUROC
$-\bar{\ell}$ (specified)	0.406	0.381
$\bar{H}$ (specified)	0.417	0.419
$-$minimum log-prob	0.660	0.697
maximum entropy	0.610	0.643
Random	0.500	0.500
Answer length only	0.972	0.971

Table 1: Task 1 on all 400 answers.

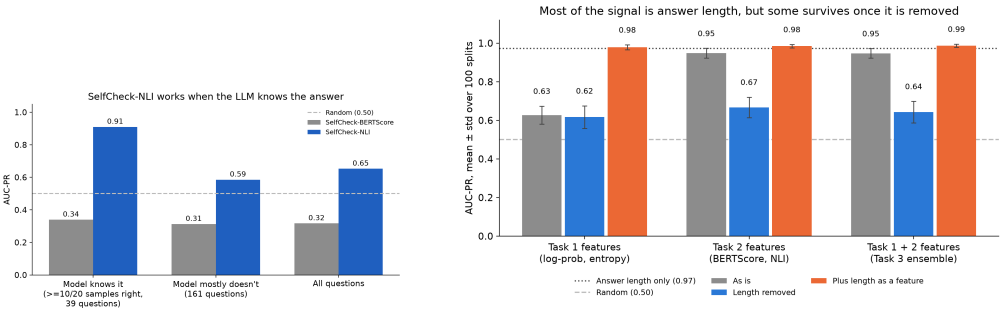
SelfCheck	AUC-PR	AUROC	Pairs
BERTScore ($n = 396$)	0.317	0.075	6%
NLI ($n = 400$)	0.653	0.614	67%
Random	0.500	0.500	50%

Table 2: Task 2 on all answers. "Pairs": fraction of questions in which the hallucinated answer receives the higher score.



(a) Task 1 (all answers) **(b)** Task 2 (all answers) **(c)** Task 3 (held-out answers)

Figure 1: Precision–recall curves. (c) shows the Task 1, Task 2 and Task 3 regressions on the same 80 held-out answers, with answer length for reference.



(a) Task 2 by model knowledge **(b)** Length control (100 splits, ±1 sd)

Figure 2: (a) AUC-PR on questions the model knows (at least 10 of 20 samples contain the correct answer) and on the remaining questions. (b) Mean AUC-PR over 100 question-level splits. Grey: raw scores; blue: length removed; orange: length added as a feature.

4.2 Task 2

- SelfCheck-NLI performs above chance (Table 2, Figure 1b). At 20% recall its precision is 0.83. Its median score is 0.920 for correct and 0.966 for hallucinated answers.
- SelfCheck-BERTScore is inverted: it assigns the higher score to the correct answer in 94% of pairs (median 0.959 for correct and 0.571 for hallucinated answers), and its sign-flipped AUC-PR is 0.918. Section 5 shows that this is caused by answer length.
- **Effect of model knowledge.** SelfCheckGPT assumes that the model generally knows the answer. Only 42% of questions have at least one sample containing the correct answer (substring match), and only 20% (39 questions) have at least 10 of 20. On these 39 questions NLI attains an AUC-PR of **0.91**, compared with 0.59 on the remaining 161 (Figure 2a). The result is stable: 0.89 and 0.93 with thresholds of 5 and 15 samples, and 0.90 after removing the 12 yes/no questions.

4.3 Task 3

- **Improvement over Task 1 and Task 2.** Task 3 attains 0.951, an improvement of +0.35 over the Task 1 regression and of +0.30 over the best raw Task 2 score (NLI) (Table 3, Figure 1c). It ranks the hallucinated answer higher in 38 of 40 test questions and classifies 87.5% of answers correctly at a threshold of 0.5.
- **No improvement over the Task 2 regression or length.** Task 3 is level with the Task 2 regression (0.957) and slightly below length alone (0.965), with overlapping intervals. Over 100 random splits: Task 1 regression 0.627 ± 0.047, Task 2 regression 0.948 ± 0.025, Task 3 0.948 ± 0.025, length 0.972 ± 0.014.

Detector (held-out 20%, $n = 80$)	AUC-PR	95% CI	AUROC
Task 1: −mean log-prob	0.429	0.40–0.51	0.422
Task 1: mean entropy	0.459	0.42–0.54	0.491
Task 1: LR(log-prob, entropy)	0.602	0.53–0.72	0.676
Task 2: SelfCheck-BERTScore	0.323	0.32–0.34	0.075
Task 2: SelfCheck-NLI	0.651	0.56–0.75	0.602
Task 2: LR(BERTScore, NLI)	0.957	0.91–0.99	0.959
Task 3: LR on all four scores	**0.951**	**0.90–0.98**	**0.951**
Task 3 + minimum log-prob, maximum entropy	0.973	0.95–1.00	0.972
Answer length only	0.965	0.94–0.99	0.969
Random	0.500	—	0.500

Table 3: Task 1, Task 2 and Task 3 on the same 80 held-out answers.

Detector (100 splits)	AUC-PR	Pairs ranked correctly
LR(length)	0.972 ± 0.014	97%
LR(length + Task 1 + Task 2)	0.987 ± 0.008	98%
LR(Task 1, length removed)	0.616 ± 0.059	74%
LR(Task 2, length removed)	0.667 ± 0.053	71%
LR(Task 1 + Task 2, length removed)	0.642 ± 0.056	75%
−mean log-prob alone, length removed	0.614 ± 0.063	75%

Table 4: Length control, mean ± standard deviation over 100 question-level splits.

- **Learned weights.** −3.08 for BERTScore, +0.99 for NLI, −0.31 for mean log-probability and −0.08 for entropy. The classifier relies mainly on BERTScore with an inverted sign. Since BERTScore is strongly correlated with log answer length ($r = -0.70$), the classifier effectively learns length.

Controlling for length. Two checks were performed, each fitted on the training answers only and repeated over 100 splits:

- **Length as an extra feature:** the regression on $\log T$ alone is compared with the regression on $\log T$ and all four scores.
- **Length removed:** each score f is regressed linearly on $\log T$ and replaced by its residual, $\tilde{f} = f - (\alpha + \beta \log T)$, and the classifier is trained on the residuals.

- Added to length, the four scores raise the AUC-PR from 0.972 to 0.987 and improve it on 95 of 100 splits (Figure 2b, Table 4).
- With length removed, the combined scores retain an AUC-PR of 0.64 and rank three of four pairs correctly. This is the most reliable estimate of what Tasks 1 and 2 contribute.
- Mean log-probability changes the most: below chance on its own, it ranks 75% of pairs correctly once length is removed, so length had biased it in the wrong direction.
- NLI falls to chance level after the correction (0.44 on the seed-42 split), so part of its advantage was also due to length.

5 Qualitative analysis

5.1 Task 1: why the mean log-probability fails

Table 5 lists the token values for one pair.

Answer	Token	$\log p$	Entropy	Top prediction
Correct	`no`	−16.30	0.00	`No`
Hallucinated	`Yes`	−15.00	0.00	`No`
	`, both documentaries`	≈ 0	low	same
	`were`	−6.84	1.19	`Tr`
	`directed by`	0.00	0.00	same
	`Johannes`	−15.64	4.84	`Liz`
	`Gren`	−10.52	1.68	`Roberts`
	`zfurthner.`	≈ 0	low	same
Mean: correct / hallucinated		−16.30 / −3.48	0.00 / 0.76	

Table 5: Pair 1290: *"Do the documentaries Traceroute and War/Dance have the same director?"* Some tokens are grouped.

- The model holds the correct answer: its top prediction for the first token is *"No"*, and all 20 samples begin with "No". However, the reference answer uses the lowercase token *"no"*, which receives a log-probability of −16.3.
- The hallucinated answer also contains two improbable tokens (*Yes*, *Johannes*), but nine near-certain tokens raise its mean to −3.48. The mean thus reflects surface form rather than factual correctness.

5.2 Task 2

Pair 7787 (correct detection by NLI)

Question: "Woh Ladki Hai Kahan", is a song, released in which year, from the movie "Dil Chahta Hai"...?
Correct: 2001 $S_{\text{NLI}} = 0.21$, $S_{\text{BERT}} = 1.06$
Hallucinated: "Woh Ladki Hai Kahan" was released in 2000. $S_{\text{NLI}} = 0.98$, $S_{\text{BERT}} = 0.42$
Qwen samples: 18 of 20 contain 2001, e.g. "...and it was released in the year 2001."

- The model knows the answer, so the samples contradict the incorrect year and NLI separates the two answers clearly.
- BERTScore ranks the pair in the wrong order, and Task 1 also fails here ($\bar{\ell} = -2.72$ for the correct answer, −1.35 for the hallucination).

Pair 117 (incorrect: BERTScore inverted)

Question: Jake David Shapiro was best know for being the screenwriter of a film directed by who?
Correct: Roger Christian $S_{\text{BERT}} = 1.10$
Hallucinated: Jake David Shapiro was best known for being the screenwriter of a film directed by Steven Spielberg. $S_{\text{BERT}} = 0.34$
Qwen samples: "Jake David Shapiro was best known for being the screenwriter of the film 'Superman II'..."

- The samples restate the question. A hallucinated answer that also restates it shares most of its tokens with them, whereas a two-word answer shares almost none, regardless of its correctness.

Pair 1169 (incorrect: the model is consistently wrong)

Question: Which film director directed the internet video serial "God, Inc.", Roman Karmen or Francis Stokes?
Correct: Francis Stokes $S_{\text{NLI}} = 0.999$
Hallucinated: Roman Karmen directed "God, Inc." with Francis Stokes. $S_{\text{NLI}} = 0.965$
Qwen samples: all 20 samples name Roman Karmen

Pair 9763 (incorrect: the samples disagree)

Question: In which city did this world games take place when Squash was added as part of the world games?
Correct: Lahti, Finland $S_{\text{NLI}} = 0.999$
Hallucinated: The World Games was never held in Lahti. $S_{\text{NLI}} = 0.990$
Qwen samples: "Toronto"; "Riga, Latvia"; "Doha, Qatar"; "... Kuala Lumpur, Malaysia in 2017."

- In pair 1169 the samples agree with one another but not with the facts, so the correct answer is flagged. In pair 9763 the samples name four different cities, so both answers are contradicted almost equally. SelfCheckGPT thus measures agreement with the model, not with the facts.

5.3 Task 3

Pair 2621 (correct: Task 1 and Task 2 agree)

Question: The 3852 Democratic National Convention nominated Franklin Pierce over what American military officer who represented Michigan...?
Correct: Lewis Cass Task 3 score 0.06
Hallucinated: James Buchanan was passed over for Franklin Pierce. Task 3 score 0.73

- In pair 2621, 7 of 20 samples name Lewis Cass. The hallucination has a lower mean log-probability (-5.96 against -2.86) and a higher NLI score (0.999 against 0.77), and both answers are classified correctly.

Pair 3657 (incorrect: labelling problem in HaluEval)

Question: ... songs by Heart, a rock band that first found success in Canada and later in the United States..., are of which nationality?
Correct: American Task 3 score 0.74
Hallucinated: Heart, the famous rock band, is American. Task 3 score 0.62

Pair 2060 (incorrect: missed hallucination)

Question: Turbidimetric inhibition immunoassay is used in many commercial immunoassays, including this medication sold under which brand name?
Correct: Lanoxin Task 3: 0.01, $S_{\text{NLI}} = 0.93$
Hallucinated: Turbidimetric inhibition immunoassay is used in many commercial immunoassays, including several medications. Task 3: 0.37, $S_{\text{NLI}} = 0.32$

- The classifier ranks the correct answer higher in 2 of 40 test questions (pairs 3657 and 6572). In both, the "hallucinated" answer is in fact correct: in pair 6572, *"... is The Chipmunks"* paraphrases *"Alvin and the Chipmunks"*. In pair 3657 all 20 samples answer "Canadian", so the model is also wrong.
- At a threshold of 0.5 there are five missed hallucinations: three are specific but incorrect (e.g. "Cancun, Mexico" instead of Daytona Beach), one nearly paraphrases the correct answer, and one makes no verifiable claim (pair 2060). An answer that avoids the question cannot be contradicted by any sample, so self-consistency cannot detect it.

6 Paper-reproduction comparison (Task 2)

- **Numbers.** For non-factual sentences on WikiBio, the SelfCheckGPT paper reports an AUC-PR of 0.925 for SelfCheck-NLI, 0.820 for SelfCheck-BERTScore and 0.832 for the average negative log-probability under GPT-3. The corresponding numbers here are 0.653 (0.910 on known questions), 0.317 and 0.406.
- **Different chance levels.** 72.96% of WikiBio sentences are non-factual, so random guessing attains 0.73 there, against 0.50 here. Relative to chance, the paper's gains are +0.20 (NLI), +0.09 (BERTScore) and +0.10 (log-probability); the gains here are +0.15 (NLI), −0.18 (BERTScore) and −0.09 (log-probability).

- **Outcome.** The gain of SelfCheck-NLI is close to the paper's, NLI again outperforms BERTScore, and on known questions the gain (+0.41) exceeds the paper's. BERTScore and the log-probability baseline do not reproduce.

Likely causes of the gap:

- **Data.** The paper scores 1,908 sentences from 238 passages written by the evaluated model (GPT-3) itself. Here, 400 answers are scored, and the hallucinations were written by ChatGPT rather than by Qwen. Qwen therefore has no particular reason to be uncertain about them, and they differ in length and style from the correct answers.
- **Compute budget.** The paper uses GPT-3 (text-davinci-003), a much larger model, on biographies of well-known people. A 3B model was used here on multi-hop questions, most of which it cannot answer (Section 4).
- **Hyperparameters and setup.** The number of samples ($N = 20$), the temperature (1.0), the BERTScore backbone and the NLI model match the paper. The main difference is the unit of comparison: the paper compares a sentence with samples of the same kind of passage, while here one-word answers are compared with full-sentence samples. This affects BERTScore most, as shown by pair 117.

7 Discussion and conclusion

What worked.

- SelfCheck-NLI is effective when its assumption holds: an AUC-PR of 0.91 on questions the model knows, without training or access to model internals.
- After controlling for length, the combined Task 1 and Task 2 scores still rank three of four pairs correctly (AUC-PR 0.64), and they add information to length (0.972 to 0.987).

What did not work.

- The two specified Task 1 scores perform below chance, and SelfCheck-BERTScore is inverted because it measures word overlap, which depends mainly on answer length.
- The Task 3 ensemble (0.951) does not improve on the Task 2 regression or on length alone, because it learns length through BERTScore.

Why.

- The hallucinations were written by a different model in a different style, so they are only weakly related to the uncertainty of the evaluated model, on which all three methods depend.
- A 3B model answering multi-hop questions without reference text usually does not know the answer. Self-consistency then gives no signal (inconsistent samples) or a misleading one (consistently wrong samples).

Limitations.

- Only 200 questions were used, so the confidence intervals are wide.
- The known-question split uses substring matching, and the length correction is linear. A length-matched test set was not feasible: only 28 pairs have answers within five tokens.

Conclusion. Token probabilities do not detect hallucinations on HaluEval QA, and SelfCheckGPT is effective only on questions the model can answer. The Task 3 score of 0.95 is largely due to answer length; with length removed, the combined scores attain 0.64.